

Name: _____

Class: _____

Date: _____

Unit 5 Vocabulary

Impact of Computing · Study Sheet



Know These Cold

Every term here can show up on the AP exam. Fold the page to hide the definitions and quiz yourself, or quiz a partner.

5.1 Beneficial and Harmful Effects

Computing innovation

includes a program as an integral part of its function (physical, nonphysical software, or a concept)

Beneficial effect

an outcome that helps people or society

Harmful effect

an outcome that hurts people, groups, or society

Unintended consequence

an effect that the creators did not plan for

Responsible programming

considering the unintended uses and effects of a program before and after release

5.2 Digital Divide

Digital divide

differing access to computing devices and the Internet

Socioeconomic

related to income, education, and social class

Geographic

related to location (rural vs. urban, country)

Demographic

related to characteristics like age, race, or gender

Equity

fairness — everyone gets what they need to succeed

5.3 Computing Bias

Bias

unfair preference for or against a group

Algorithmic bias

bias that results from an algorithm's design or its data

Training data

the data a machine learning model learns from

Machine learning

computers learning patterns from data instead of explicit rules

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Representative data

data that reflects the whole population it will be used on

5.4 Crowdsourcing

Crowdsourcing

getting input or information from many people via the Internet

Citizen science

research done with help from non-scientists who contribute data

Crowdfunding

raising money from many people online

Public data

data made freely available for anyone to use

Collaboration

people working together toward a common goal

5.5 Legal and Ethical Concerns

Intellectual property

creations of the mind — code, art, music, writing — owned by the creator

Copyright

legal protection that gives creators control over how their work is used

Creative Commons

a license creators use to give others the right to share, use, and build upon their work

Open source

code that can be freely shared and modified

Plagiarism

using someone else's work without permission and presenting it as your own

5.6 Safe Computing

Encryption

the process of encoding data to prevent unauthorized access

Symmetric key

encryption that uses one key for both encrypting and decrypting

Public key

pairs a public key for encryption with a private key for decryption

Certificate authority

issues digital certificates that validate ownership of encryption keys (a trust model)

Malware

software intended to damage a computing system or take partial control of it